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Evaluating Brayton Heat Pump Potential for Industrial Decarbonisation

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Abstract. High-temperature heat pumps facilitate the integration of renewable into industrial processes and foster thermal production decarbonisation in industry. Among the various options available, Brayton heat pumps are increasingly researched due to their ability to produce heat at higher temperatures than the vapour-compression alternatives, for which commercial solutions are limited to up to 200 °C. While Brayton HTHPs feature a maximum heat production temperature that may easily overcome 350 °C, this technology does not produce heat at a constant temperature, such that the average temperature of the produced heat may be much lower than its maximum value and more in line with vapour compression technologies, potentially vinifying the competitive advantage of Brayton HTHPs. This paper reviews the applicability fields of Brayton HTHP, considering the maximum and minimum temperature values required by various industrial processes. Furthermore, the trade-off between the COP and the average temperature of the produced heat is explored for various Brayton HTHP configurations by considering the practical limitations that operating pressures, temperatures and mass flow rates impose on the HTHP components, namely compressors, expanders and heat exchangers. In this way, a comprehensive characterisation of the Brayton HTHP technology and its potential for industrial decarbonisation is assessed.

1. Introduction

1.1 Literature Background

Fossil fuels are the industry's primary source of heat generation, and the related decarbonisation process may pose significant technical, economic, and societal issues [1]. One proposed strategy for heat production decarbonisation relies on electrification, which can be done by introducing into industrial processes electric heaters/boilers or High-Temperature Heat Pumps (HTHP), often combined with mechanical vapour recompression [2].

Focusing on HTHP, they are becoming a promising option for heat production in the chemical, food, metal, and paper industries [3], and several practical applications have already been successfully demonstrated [4]. Seven of the 18 case studies collected by [4] are dedicated to drying applications in different industries, while the others are used to produce steam at medium-low temperatures (less than 200 °C). Economic and technical barriers currently limit HTHP's widespread adoption [5], like the inability to operate efficiently above 200 °C and with temperature lifts above 100 K [6], even though several industrial sectors show a potential demand

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for HTHP operating up to these or even higher ranges [7]. The limitation on the maximum operating temperature is linked to the compressor lubrication and thermal control. While 180 °C [5,6] was considered the upward limitation just a few years ago, commercial products operating up to 200 °C are now being proposed [4].

The above mentioned limitations characterise the most mature HTHP architectures, i.e., the vapour compression systems [4], but do not affect HTHP based on reverse Brayton cycles. Brayton HTHP features lower COP than vapour compression technologies for similar temperature lifts, but they may operate at temperatures far beyond those accessible by the former. For example, [8] investigated a Brayton HTHP, finding that a COP comprised between 1.65 and 1.75 is expected while operating up to 320 °C, with a lift of 260 K. The system studied in the paper is currently being commissioned at the DLR facility in Cottbus and is expected to operate soon, demonstrating the Brayton HTHP technology [9].

Besides the very high operating temperatures, Brayton HTHPs use safe, non-toxic, non-flammable, potentially inexpensive and climate-neutral operating fluids (such as Air, Argon, Nitrogen and CO_2). In particular, operating with Air unlocks open cycles, which operate at pressures in the orders of a few bars, much lower than those commonly found in vapour compression applications. When closed cycles are operated, Brayton HTHP can achieve very extended part-load flexibility due to inventory control [10], which allows changing the thermal production by simply manipulating the gas mass contained in the cycle volume (by extracting or injecting mass). This can be done without altering the cycle temperatures and pressure ratios, such that the COP remains unaltered even for significant load variations. In [10], the authors simulate the part load performance of a Brayton HTHP equipped with inventory control, finding that a 25 % variation in the produced heat at the sink corresponds to a COP increase of 8%. In contrast, in analogy with their low-temperature counterparts, vapour compression technologies are expected to experience around 10% COP reduction for similar load reductions [11,12].

Brayton HTHP produces sensible heat, making it the best option when steam is not requested, like producing hot air in various applications. The industrial processes that require a large share of heat at temperatures between 200 °C and 500 °C are secondary aluminium production and rolling/extruding of aluminium, flat glass, gypsum, and ethylene production [13]. In [14], four industrial case studies from the food and chemical industries are simulated, comparing Brayton, vapour compression, and transcritical CO2 HTHPs. As expected, even for relatively low temperatures (up to 150 °C), Brayton HTHP outperformed the other technologies when both source and sink provided and requested heat at various temperature levels.

Regardless of the heat (latent or sensible) type provided and requested in a process, Brayton HTHP is the most developed option for producing heat at temperatures beyond 150 – 200 °C. Despite this, Brayton HTHPs may still struggle to achieve financial viability [13].

To facilitate Brayton HTHP's widespread adoption, it is helpful, on one side, to focus on simple, compact, and cheap layouts, which means, for example, operating with open cycles, eliminating the need for a heat exchanger between the cycle fluid and the environment. For heat exchangers, this implies limiting the dimensions of the components, adopting realistic thermal effectiveness up to 0.8 [8]. The internal heat exchanger (IHX) plays a crucial role, allowing internally regenerated Brayton HTHP to achieve very high temperatures with low-pressure ratios. However, by limiting the IHX size excessively, the cycle of COP can be deeply reduced [15].

Another way to foster Brayton HTHP applications is to develop systems that may operate satisfactorily up to 300 – 350 °C and above, a temperature range with currently no technological

options besides electric heaters. Compared to direct electric heating, which has COP = 1, even operating with COP in the order of 1.2 – 1.3 would significantly reduce energy consumption.

1.2 Study Innovative Contributions

To the best of the authors' knowledge, the current research on Brayton HTHP lacks a clear understanding of the impact that cycle operating conditions and layout have on the COP and the temperatures at which the heat is produced. In this landscape, what is usually done is to evaluate the COP and the maximum achievable sink temperature without explicitly gaining clear insights into the average temperature at which this heat is produced and how this affects the COP. Furthermore, all the available studies focus exclusively on the simple internally regenerated cycle without considering the possibility of adding simple yet impactful layout modifications such as inter-refrigerated compression.

A research gap is to be covered in this regard: to map COP and average heat production temperature $\bar{T}_{Mean}$ in function of the cycle operating conditions and new layouts. The present paper contributes to the research on Brayton HTHP by proposing the following innovative elements:

- A multi-criteria analysis of the Brayton HTHP thermodynamic cycle is proposed, systematically investigating the trade-off between COP and average heat production temperature for the first time;
- The paper introduces the possibility of using an inter-refrigerated compression and demonstrates its usefulness in improving the average heat production temperature. While this cycle layout has already been investigated for low-temperature applications [16], the paper will focus specifically on how the inter-refrigerated compression impacts the size of the internal heat exchanger (IHX), which has never been evaluated before, to the best of the authors' knowledge.

2. Methodology

This study analyses an open Brayton cycle with inter-cooled compression and internal regeneration. In Figure 1, the considered cycle layout and T-s diagram are reported.

The equations used are the energy balance that allows the definition of the entire cycle once some parameters have been set. In Table 1, the chosen values for these parameters are reported together with the range of variation of those parameters that have been allowed to vary.

In Table 1, $split$ represents how much the total pressure ratio of the cycle (β_{Tot}) is distributed between the first and the second compressors, which are assumed to have the same isentropic efficiency (η_{Cmp}^{Is}). Hence, the pressure ratio of each of the two compressors can be expressed as follows:

$$\begin{aligned} \beta_1 &= \beta_{Tot}^{1-spl} \\ \beta_2 &= \beta_{Tot}^{spl} \end{aligned} \qquad (1)$$

β_1 and β_2 are symmetric with respect to $spl = 0.5$, which corresponds to $\beta_2 = \beta_1$. Varying $split$ from 0 to 1 allows the exploration of all the solutions from the case where there is only the first compressor to the case where there is only the second one.

The mass and heat balance equations on each cycle component have been used to evaluate the cycle COP, which can be expressed as follows (refer to Figure 1 for the formula interpretation):

$$COP = \frac{h_{b''} - h_c + h_b - h_{b'}}{h_b - h_a + h_{b''} - h_{b'} - (h_d - h_e)} \tag{2}$$

Enthalpies can be expressed as functions of the parameters in Table 1 with the addition of the internal regenerator's effectiveness (ε_{ihx}) and since exists a unique value of ε_{ihx} such that $T_f = T_{Env}$ (as it should be in an open cycle) ε_{ihx} can be found iteratively by imposing this condition. Moreover, having the ε_{ihx} as an output allows us to understand the impact that the other imposed parameters have on heat exchanger effectiveness and the entire cycle performance.

Once the entire cycle is specified, other relevant parameters can be evaluated, like the average heat production temperature, defined as the average of the two mean temperatures at which heat is delivered to the sink in the intercooler and the cooler, weighted by the amount of heat transferred in each heat exchanger:

$$\bar{T}_{Inter} = \frac{h_b - h_{b'}}{s_b - s_{b'}} \tag{3}$$

$$\bar{T}_{Cooler} = \frac{h_{b''} - h_c}{s_{b''} - s_c} \tag{4}$$

$$T_{ave} = \frac{\bar{T}_{Inter} \cdot (h_b - h_{b'}) + \bar{T}_{Cooler} \cdot (h_{b''} - h_c)}{h_{b''} - h_c + h_{b''} - h_c} \tag{5}$$

In both cases, weighting the average heat production temperature in the cooler and the intercooler against the total heat transferred to the sink prevents favouring solutions where the average temperature is very high, but the exchanged heat is insignificant.

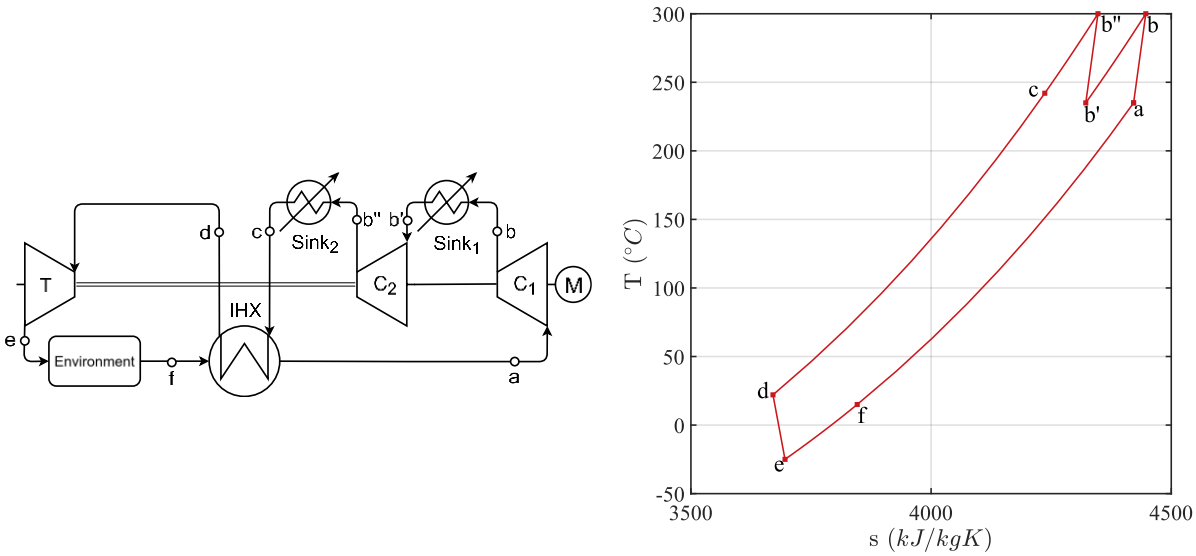


Figure 1 – Cycle layout (left) and T-s diagram (right) of an open Brayton cycle with inter-cooled compression and internal regeneration. The "f" point represents the environmental conditions. $T_f = 15$ °C.

Parameter	η^{Is}_{Cmp} (-)	η^{Is}_{Trb} (-)	T_{Env} (°C)	T_{Max} (°C)	T_{Min} (°C)	β_{Tot} (-)	$split$ (-)
Cycle	0.8	0.9	15	300	-50 ÷ 0	1.1 ÷ 5.0	0 ÷ 1.0

Table 1 – Values of set and varied parameters

3. Results

By varying the minimum temperature of the cycle and the *split* for a constant maximum temperature, the behaviour of different solutions can be explored, as reported in Figure 2, in terms of COP and average heat production temperature.

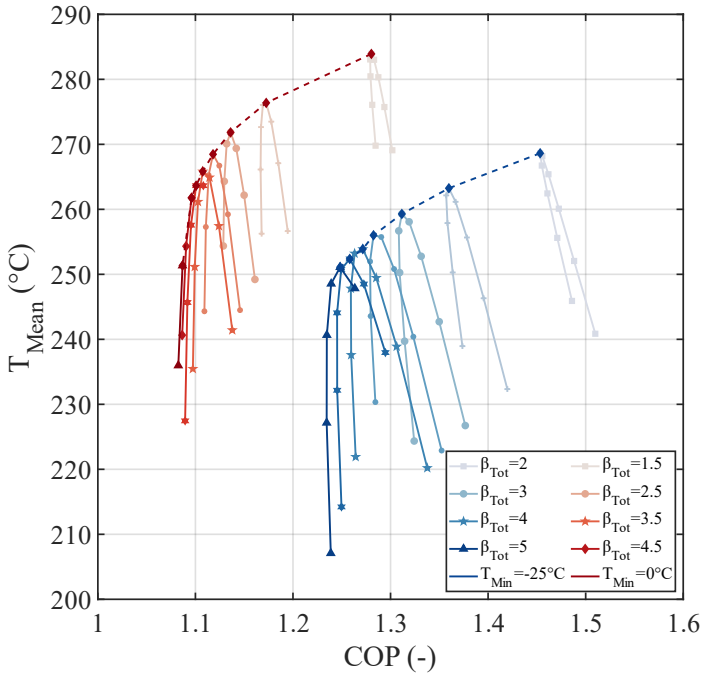


Figure 2 – Trend of average heat production temperature with COP for different values of the minimum temperature of the cycle and different total pressure ratios.

Here, two different groups of curves are represented for different values of T_{Min}, and the total pressure ratio increases while moving from one curve to another from right to left for each T_{Min}. Finally, each curve has as many points as the number of *split* values simulated for it. If a curve has fewer points than the others, it means that not all combinations of the parameters T_{Max}, T_{Min}, β_{Tot}, and *split* yield physically feasible solutions.

From Figure 2, it is observed that, for each value of T_{Min}, the optimal solution in terms of COP and T_{Mean} is the one with the minimum pressure ratio. This behaviour is closely linked to the effect that the pressure ratio has on T_{Mean} and COP.

As the compression ratio increases, the first compressor inlet temperature decreases for fixed maximum and minimum cycle temperatures, consequently lowering the inlet temperature to the regenerator. The direct result of this is a decrease in T_{Mean} (since T_{Max} is fixed) and increased heat transfer per unit mass to the hot source. At the same time, since T_{Min} and the ambient temperature are fixed, the heat absorbed per unit mass from the cold source remains constant, thereby causing a reduction in the COP, as reported in Figure 3a.

Moreover, the *split* that features the higher COP is 0 (or 1, not reported in the plot since it behaves the same as 0). In this case, there is only one compressor and no inter-refrigeration. Even in this case, the reduction in the COP is associated with increased heat transferred to the hot source per unit mass while keeping the heat absorbed from the cold source per unit mass constant.

On the other hand, the split value that maximizes T_{Mean} decreases with β_{Tot} as reported in Figure 3b. This is because T_{Mean}, being calculated as a weighted average, represents a trade-off between $\bar{T}_{Inter}$, $\bar{T}_{Cooler}$, and the heat exchanged in the cooler and intercooler (see Equation (5)).

Furthermore, at a constant split, an higher β_{Tot} leads to a lower compressor inlet temperature. Therefore, as T_a also lower the regenerator inlet temperature, a high β_{Tot} causes T_c to decrease more swiftly with the split, thereby penalizing T_{Mean}.

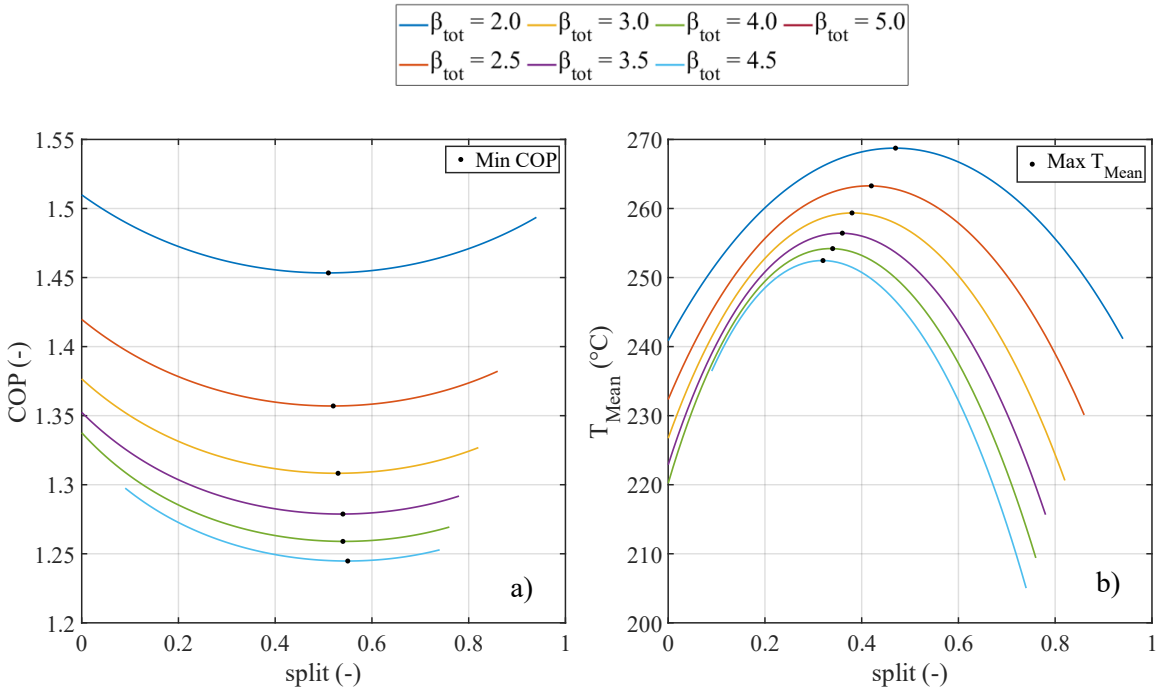


Figure 3 –COP (a) and T_{Mean} (b) for various split values. $T_{Max} = 300°C$ and $T_{Min} = -25°C$.

3.1 Multi-criteria analysis

Based on all the considerations discussed above, we can deduce that the practical solutions in terms of the parameters T_{Min}, β_{Tot}, and $split$ are those within the range from split=0 to the value that maximizes T_{Mean}. This is because solutions with split values larger than the one that maximizes T_{Min} feature both lower COP and lower T_{Mean}. Thus, we can represent a type of Pareto front that showcases all the solutions of practical interest in Figure 4.

Hence, from Figure 4, the most suitable solution can be chosen depending on whether the focus is more on achieving a high average heat production temperature or a high COP.

However, when choosing the appropriate combination of $split$, β_{Tot} and T_{Min} carefull attention must be given to the requested effectiveness of the regenerator. From Figure 4 it can be seen how the highest values of both COP and T_{Mean} required values of ε_{IHX} which are not technically feasible for air-to-air heat exchangers. Thus, if an effectiveness of 0.8 is taken as in [17], results with T_{Mean} greater than 275°C or COP greater than 1.4 are practically non feasible.

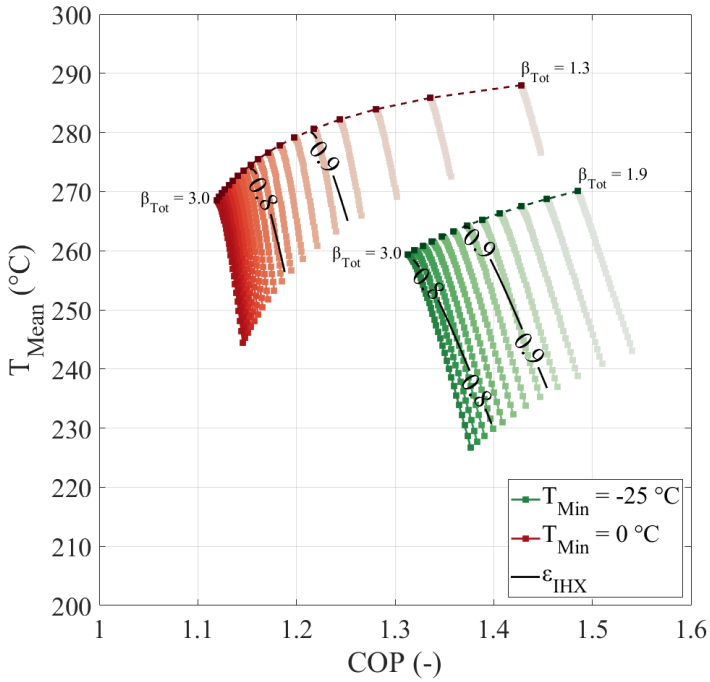


Figure 4 – Different Pareto fronts for different combinations of T_{Min}, β_{Tot} and split with T_{Max} fixed at 300°C

3.2 Cycle domain of feasibility

As it resulted from Figure 2, the higher T_{Min} the higher T_{Mean}, while the lower T_{Min} the higher COP. Furthermore, it is worth noticing that when β_{Tot} is low, the maximum value of T_{Mean} correspond to the minimum value of COP, whereas for higher values of β_{Tot} the maximum value of T_{Mean} is characterised by a COP value slightly better than the minimum for that pressure ratio.

As mentioned before, not all the combinations of T_{Max}, T_{Min}, β_{Tot}, and $split$ yield physically feasible solutions. Hence it is interesting to define the set of feasible thermodynamic cycles for different values of T_{Min}, β_{Tot}, and $split$ as reported in Figure 5.

The effectiveness of the regenerator increases with the $split$, while it decreases with the pressure ratio. This means that since the values that maximize T_{Mean} and COP are achieved at a low β_{Tot}, regenerators with very high effectiveness, which implies extensive heat transfer areas and high costs, are required. In particular, the solution that maximizes T_{Mean} requires even higher effectiveness compared to the one maximizing the COP because for the same β_{Tot}, it has a higher $split$.

The reason why the $split$ and the regenerator effectiveness increase together is because the first compressor inlet temperature grows when the split increases. As shown in Figure 6, which depicts the T-s diagrams for different splits but the same β_{Tot}, as the $split$ increases, T_a also increases, which in turn causes an increase in T_c, while T_d and T_f remain constant. In a heat exchanger, in general, an increase in the inlet temperature of the hot fluid while keeping the outlet temperature of the fluid and the inlet temperature of the cold fluid constant leads to an increase in effectiveness, as observed in this case.

Another peculiarity that can be observed in Figure 5 is how the shape of the domain of feasibility varies alongside T_{Min}. In particular, the domain of feasibility tends to shorten and widen as T_{Min} decreases. In other words, for lower values of T_{Min}, the domain of feasibility becomes wider for the same β_{Tot}, but it is limited on the lower side by higher values of β_{Tot} and this has

implications for system costs. Specifically, for the same T_{Max}, achieving higher COP requires reaching a lower T_{Min}values. However, to achieve this, it is necessary not only to have high ε_{ihx} values, which entail high costs, but also higher β_{Tot} values, which will also tend to increase costs.

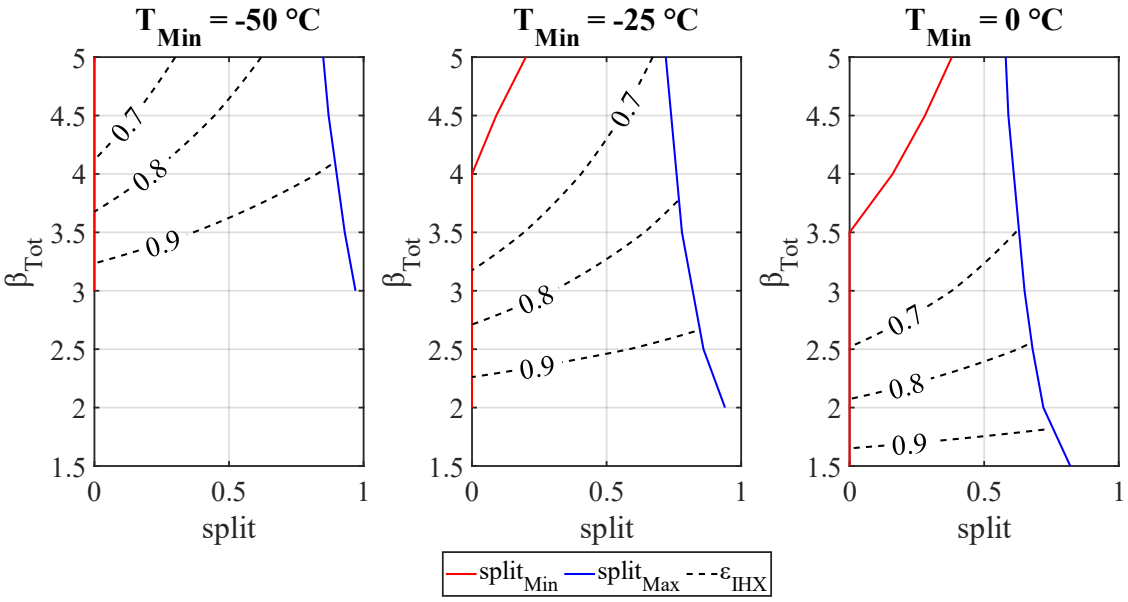


Figure 5 – Feasibility domain for different values of T_{Min}

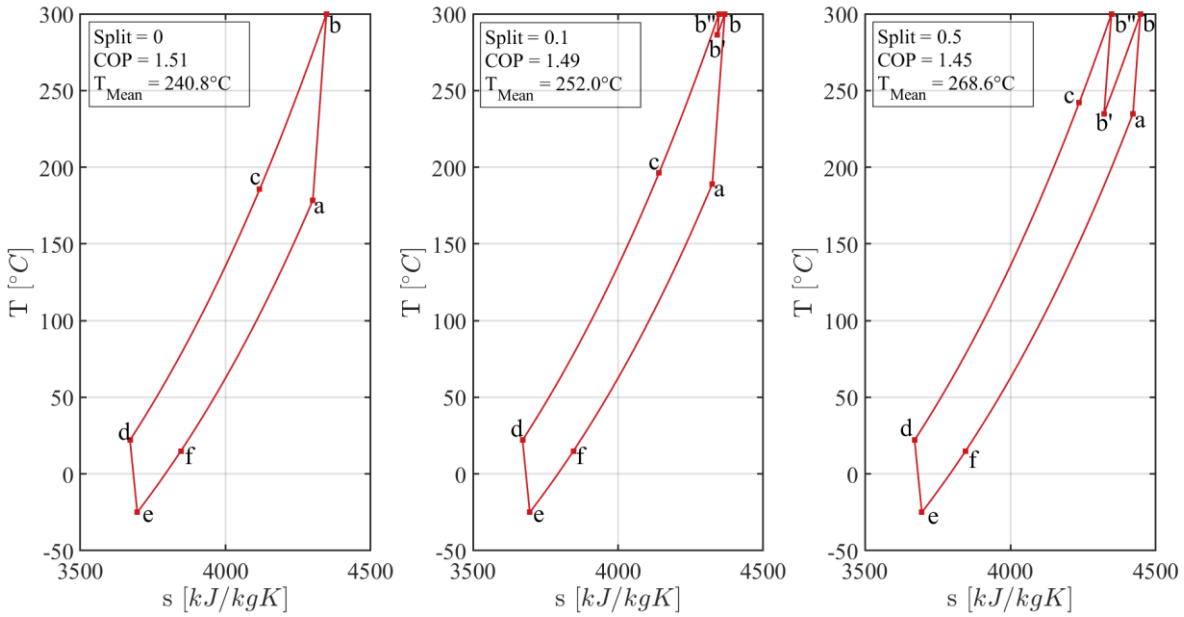


Figure 6 – T-s diagram for different values of split with constant pressure ratio ($\beta_{Tot} = 2$) and constant minimum and maximum temperatures ($T_{Max} = 300°C; T_{Min} = -25°C$).

4. Conclusions

This paper investigates the impact of inter-cooled compression with internal regeneration on the COP and average heat production temperature when the heat from the inter-cooled compression is recovered and used to improve the average heat production temperature. To achieve this, a parameter called *split* has been introduced to account for the distribution of the pressure ratio between the two compressors.

The results show that the higher the minimum temperature of the cycle, the higher the average heat production temperature, whereas the lower the minimum temperature of the cycle, the higher the COP. However, for a fixed value of minimum and maximum temperature of the cycle, an increase in the pressure ratio results in a decrease in both the COP and the average heat production temperature.

Moreover, the *split* plays a crucial role in determining the values of both the COP and the average heat production temperature. Specifically, the *split* that maximizes the COP is always 0, indicating that the best solution for the COP is without inter-cooling. Conversely, from the perspective of average heat production temperature, there is a specific *split* that maximizes it. Unlike the *split* that maximize the COP, which does not vary if the other parameters change, the values that maximize the average heat production temperature decrease while the pressure ratio is increasing and the minimum temperature of the cycle decreases.

Finally, it is worth noting that to maximize both the average heat production temperature and the COP, small pressure ratio values should be used. However, in these cases, high values of internal regenerator effectiveness are required, which leads to practical and feasible solutions with a maximum T_{Mean} value of 275°C and a COP of 1.15 or a maximum COP of 1.36 with a T_{Mean} of 225°C.

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References

[1] Manuel SD, Floris T, Kira W, Jos S, André F. High technical and temporal resolution integrated energy system modelling of industrial decarbonisation. Adv Appl Energy 2022;7:100105. doi:10.1016/j.adapen.2022.100105.

[2] Bless F, Arpagaus C, Bertsch SS, Schiffmann J. Theoretical analysis of steam generation methods - Energy, CO 2 emission, and cost analysis. Energy 2017;129:114–21. doi:10.1016/j.energy.2017.04.088.

[3] Arpagaus C, Bless F, Uhlmann M, Schiffmann J, Bertsch SS. High temperature heat pumps: Market overview, state of the art, research status, refrigerants, and application potentials. Energy 2018;152:985–1010. doi:10.1016/j.energy.2018.03.166.

[4] IEA. HPT TCP Annex 58 Task 1: Technologies - State of the art and ongoing developments for systems and components. 2023.
[5] Schlosser F, Jesper M, Vogelsang J, Walmsley TG, Arpagaus C, Hesselbach J. Large-scale heat pumps: Applications, performance, economic feasibility and industrial integration. Renew Sustain Energy Rev 2020;133:110219. doi:10.1016/j.rser.2020.110219.
[6] Bergamini R, Jensen JK, Elmegaard B. Thermodynamic competitiveness of high temperature vapor compression heat pumps for boiler substitution. Energy 2019;182:110–21. doi:10.1016/j.energy.2019.05.187.
[7] Marina A, Spoelstra S, Zondag HA, Wemmers AK. An estimation of the European industrial heat pump market potential. Renew Sustain Energy Rev 2021;139:110545. doi:10.1016/j.rser.2020.110545.
[8] Gollasch J, Lockan M, Stathopoulos P, Nicke E. Multidisciplinary Optimization of Thermodynamic Cycles for Large-Scale Heat Pumps With Simultaneous Component Design. J Eng Gas Turbines Power 2024;146:1–35. doi:10.1115/1.4063637.
[9] DLR. Co-Bra sup-ports the ther-mal tran-si-tion in in-dus-try 2023. https://www.dlr.de/en/latest/news/2022/03/cobra-supports-the-thermal-transition-in-industry.
[10] Pettinari M, Frate GF, Tran AP, Oehler J, Stathopoulos P, Kyprianidis K, et al. Impact of the Regulation Strategy on the Transient Behavior of a Brayton Heat Pump. Energies 2024;17:1020. doi:10.3390/en17051020.
[11] Bischi A, Taccari L, Martelli E, Amaldi E, Manzolini G, Silva P, et al. A detailed MILP optimization model for combined cooling, heat and power system operation planning. Energy 2014;74:12–26. doi:10.1016/j.energy.2014.02.042.
[12] Mazzoni S, Ooi S, Nastasi B, Romagnoli A. Energy storage technologies as techno-economic parameters for master-planning and optimal dispatch in smart multi energy systems. Appl Energy 2019;254:113682. doi:10.1016/j.apenergy.2019.113682.
[13] Zühlsdorf B, Bühler F, Bantle M, Elmegaard B. Analysis of technologies and potentials for heat pump-based process heat supply above 150 °C. Energy Convers Manag X 2019;2:100011. doi:10.1016/j.ecmx.2019.100011.
[14] Gai L, Varbanov PS, Walmsley TG, Klemeš JJ. Critical Analysis of Process Integration Options for Joule-Cycle and Conventional Heat Pumps. Energies 2020;13:635. doi:10.3390/en13030635.
[15] White AJ. Thermodynamic analysis of the reverse Joule–Brayton cycle heat pump for domestic heating. Appl Energy 2009;86:2443–50. doi:10.1016/j.apenergy.2009.02.012.
[16] Wang S, Li S, Jiang S, Wu X. Analysis of the Air-Reversed Brayton Heat Pump with Different Layouts of Turbochargers for Space Heating. Buildings 2022;12:870. doi:10.3390/buildings12070870.
[17] Walden JVM, Bähr M, Glade A, Gollasch J, Tran AP, Lorenz T. Nonlinear operational optimization of an industrial power-to-heat system with a high temperature heat pump, a thermal energy storage and wind energy. Appl Energy 2023;344:121247. doi:10.1016/j.apenergy.2023.121247.